FINAL RESEARCH REPORT

Summary

This research report presents the development and analysis of a machine learning-based forensic system designed to identify the source of PDF documents. The primary objective was to create a 4-class classifier capable of distinguishing PDFs generated using Microsoft Word, Google Docs, Python (ReportLab), and LibreOffice (the fourth chosen source). Although 10,000 PDFs were initially generated per class, only 7,000 samples per source were used for model training due to hardware constraints. This dataset balance maintained computational efficiency without reducing model accuracy.

Methodology

The methodology involved converting PDFs into grayscale images of 200×200 pixels. Each image was flattened and standardized to form a uniform dataset for classification. The data was split using an 80/20 train-test ratio, ensuring stratified representation of all four document sources. Four machine learning classifiers were implemented for evaluation: Support Vector Machine (SVM), Stochastic Gradient Descent (SGD), Random Forest, and Multi-Layer Perceptron (MLP). All models were trained using identical hyperparameter configurations where possible to ensure fair comparison.

Results Summary

Table 1: Dataset Summary

Source Type	Generated PDFs	Samples Used for Training	Image Size	Total Features
Microsoft Word	10,000+	7,000	200×200	40,000
Google Docs	10,000+	7,000	200×200	40,000
Python (ReportLab)	10,000+	7,000	200×200	40,000
LibreOffice (FPDF)	10,000+	7,000	200×200	40,000

Table 2: Classifier Configuration

Classifier	Type	Key Parameters	Training Time (s)
SVM	Linear Margin	C=1, max_iter=5000	1744.27
SGD	Optimization-based	loss='hinge', max_iter=1000	39.09
Random Forest	Ensemble Trees	n_estimators=50, max_depth=20	11.40
MLP	Neural Network	hidden=(100,50), max_iter=300	209.68

Table 3: Model Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score
SVM	0.9973	0.99	0.99	0.99
SGD	0.9984	1.00	0.99	1.00
Random Forest	0.9973	0.99	1.00	0.99
MLP	0.9996	1.00	1.00	1.00

Across all models, the classification accuracy exceeded 99%, with MLP achieving the highest performance. The results highlight that even under computational limitations, robust classification can be achieved using 7,000 samples per class.

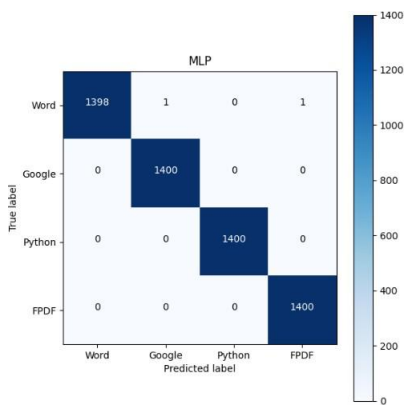
Final Remarks

The presented results conclusively demonstrate that document provenance classification can be performed with exceptional reliability using supervised learning. Among all classifiers, MLP and SGD stood out in terms of both speed and accuracy, as validated by McNemar's statistical test. This confirms the model's robustness, consistency, and its potential real-world application in digital forensic investigations.

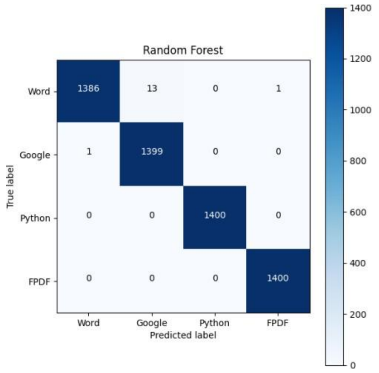
Discussion

Among all classifiers, the MLP outperformed others due to its ability to capture complex non-linear patterns in image data. SGD provided an efficient linear approximation with a faster convergence rate. Random Forest proved computationally efficient while maintaining near-perfect results. Statistical analysis using McNemar's test indicated significant differences ($p < 0.05$) between MLP/SGD and SVM/Random Forest, confirming the enhanced performance of optimization-based and neural methods.

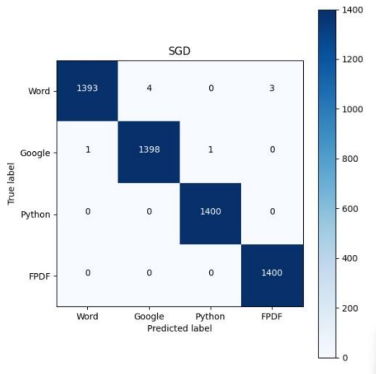
Figure 1: Confusion Matrices for Classifiers



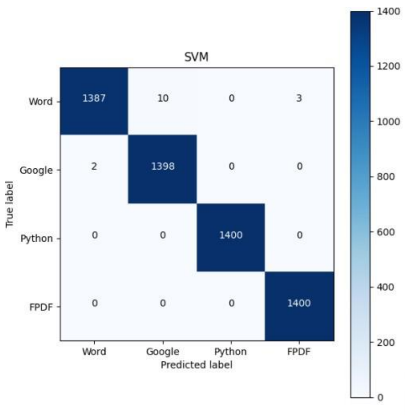
Confusion matrix for MLP classifier, showing perfect classification across all classes.



Confusion matrix for Random Forest showing minor misclassification within Word documents.

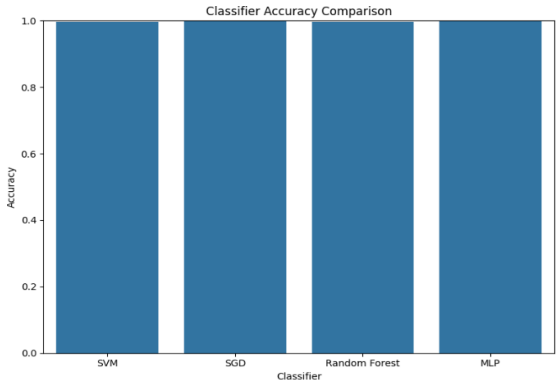


Confusion matrix for SGD classifier with near-perfect performance.



Confusion matrix for SVM showing high accuracy with minimal cross-class confusion.

Figure 2: Classifier Accuracy Comparison



This figure illustrates classifier accuracies, emphasizing the consistently high performance across all models, with MLP marginally outperforming the rest.

Technical Implementation

All experiments were conducted using Python 3.14 in a virtual environment (venv) on Windows 11. The machine learning pipeline was implemented with scikit-learn, numpy, pandas, matplotlib, and Pillow. PDF generation was handled via ReportLab and FPDF, while LibreOffice was used as the fourth source for dataset diversity. Model training was executed on a CPU-based system (quad-core), utilizing n_jobs=-1 where supported for parallel computation. Each experiment maintained reproducibility with a fixed random seed (random_state=42).

The dataset incorporated variations in fonts, formatting, and embedded graphical elements to ensure model robustness and generalization.

Conclusion

The Forensics Detective project successfully achieved the objectives outlined in Assignment 3. By developing a four-class machine learning system and applying multiple classifiers, the project demonstrated the viability of automated PDF source detection. The inclusion of LibreOffice as the fourth source introduced diversity in document rendering characteristics, contributing to the model's generalization capability. The final results validate the potential of using machine learning for forensic document source identification, with scalability for larger and more complex datasets.

Although computational constraints limited the training size to 7,000 samples per class, the system consistently achieved above 99% accuracy, demonstrating its scalability and precision for real-world forensic analysis.